

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
7th Semester of B. Pharm Examination
Nov-Dec 2015

PH403 Medicinal Chemistry–III

Date: 07.12.2015, Monday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A Give the chemical classification of antihypertensive agents & explain SAR of 2,4-dihydropyridine derivatives as calcium channel blockers in details. 10
- B Write a brief note on docking process with their application in modern drug discovery. 10
- C I. Write a classification of drugs used as anti arrhythmic agents. 05
II. Explain SAR of ACE Inhibitors. 05

Q 2 Attempt any TWO of the following

- A I. Write a short note on anticoagulants. 05
II. Write short note on anti angina agents. 05
- B I. Explain SAR of thiazide diuretics. Write a synthetic pathway of furosamide. 05
II. Explain SAR & Mechanism of action of HMG Co-A reductase inhibitors. 05
- C I. Explain requirements for docking process. 05
II. Differentiate structure based drug design & ligand based drug design. 05

SECTION- II

Q 3 Attempt any TWO of the following

- A Enlist the physicochemical parameters used in QSAR studies. Explain taff's steric constant, hammet constant in brief. 10
- B What is CoMFA & CoMSIA? Explain Methodology of CoMFA. 10
- C I. Explain in Hansch Analysis in details. 05
II. Write a short note on Topliss scheme for aromatic and aliphatic substitutions. 05

Q 4 Attempt any TWO of the following

- A Explain Following terms: Lead, Hit, High Throughput Screening, Random Screening & New uses for old drug approach. 10
- B I. Write a note on analogue approach for lead identification. 05
II. Write a short note on SOSA with suitable examples. 05
- C I. Differentiate 2D QSAR & 3D QSAR. 05
II. Enlist different model used for QSAR studies. Explain Crag's Plot in details. 05